

MAI 5301 - Week 15 Presentation

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GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models

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Background & Motivation

Rapid growth of generative AI, especially LLMs

LLMs can perform tasks across many domains (text, code, analysis)

Not limited to one industry or function

Raising important questions about impact on jobs

Need to understand how work may change

Exam results (ordered by GPT 3.5 performance)

Estimated percentile lower bound (among test takers)

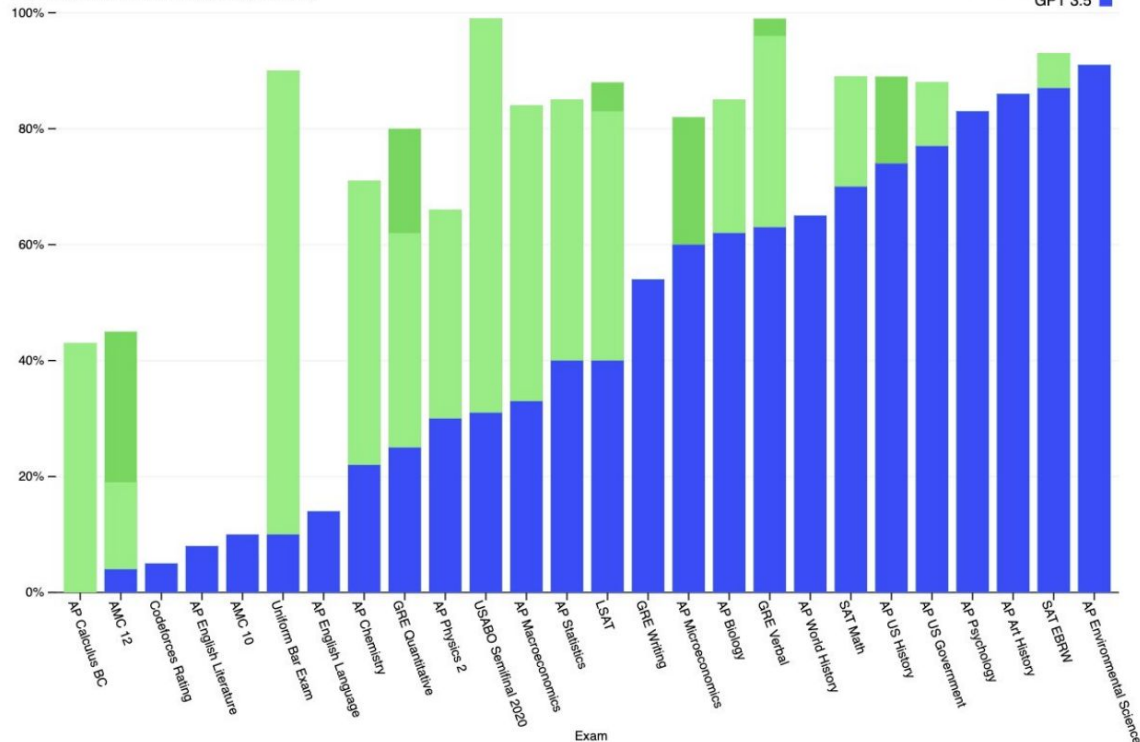


Figure 1: Taken directly from GPT-4 Technical Report (OpenAI, 2023b). To get a sense of how quickly model capabilities are progressing – consider the jump in exam performance between GPT-3.5 and GPT-4 (OpenAI, 2023b).

What This Study Focuses On

Focus on text and code-based LLMs (e.g., GPT-type models)

Includes both LLMs and LLM-powered software tools

Studies impact at the task level, not just job level

Key idea: jobs are made up of many tasks

Some tasks are more affected than others

Core Research Concept – “Exposure”

Exposure = how much LLMs can reduce task time ($\geq 50\%$)

Measures potential efficiency gains, not job loss

Does NOT mean full automation or replacement

Captures both direct use and tool-assisted use

Helps compare impact across jobs

Research Approach & Contribution

Uses O*NET dataset of occupations and tasks

Combines human judgment + GPT-4 evaluation

Introduces a new rubric for measuring LLM impact

One of the first large-scale studies focused on LLMs

Provides systematic, data-driven insights

Literature Review - Evolution of LLMs

Rapid improvements in AI capabilities in recent years

LLMs perform writing, coding, translation, and reasoning tasks

Driven by more data, larger models, better training

Increasingly accurate and easier to use

Expanding into real-world applications

From Tools to Systems

Earlier AI tools handled single tasks

LLMs act as flexible, multi-purpose systems

Can integrate with APIs, search tools, and software

Enable more complex workflows

Moving toward general-purpose digital assistants

Role of Complementary Technologies

LLMs alone have limitations (errors, bias, reliability)

Tools built around LLMs increase usefulness

Examples: coding assistants, research tools, automation systems

Human oversight often still required

System integration drives real-world value

AI and the Labor Market

Jobs consist of many different tasks

Technology affects tasks, not entire jobs at once

Routine tasks are easier to automate

Can reshape wages and job structures

Known as task-based or “routine-biased” change

Measuring AI Impact in Past Research

Uses task databases like O*NET

Applies expert judgment and machine learning

Includes “Suitability for Machine Learning” measures

Some focus on full job automation risk

This study extends these methods to LLMs

General-Purpose Technologies (GPTs)

Technologies that affect many industries (e.g., electricity)

Improve over time and enable new innovations

Spread slowly but widely across the economy

Require complementary systems to unlock full value

LLMs may fit this category

Methods and Data - Data Sources

O*NET database: 1,000+ occupations, 19,000+ tasks

Includes Detailed Work Activities (DWAs)

Combined with labor data (wages, education, employment)

Focus on U.S. workforce

Links tasks to real-world jobs

Exposure Framework

Exposure = $\geq 50\%$ reduction in task time

Focus on technical potential, not actual use

Measures efficiency gains from AI

Does not distinguish augmentation vs automation

Standardizes comparison across jobs

Exposure Categories & Metrics

E0: no impact | E1: direct LLM impact | E2: tool-assisted impact

α (alpha): direct LLM exposure

β (beta): combined exposure (main measure)

ζ (zeta): full exposure including all tools

Captures increasing levels of impact

Evaluation Method

Tasks labeled by human annotators

GPT-4 used to classify tasks with structured prompts

High agreement between human and AI results

Strengthens reliability of findings

Demonstrates LLMs can evaluate work tasks

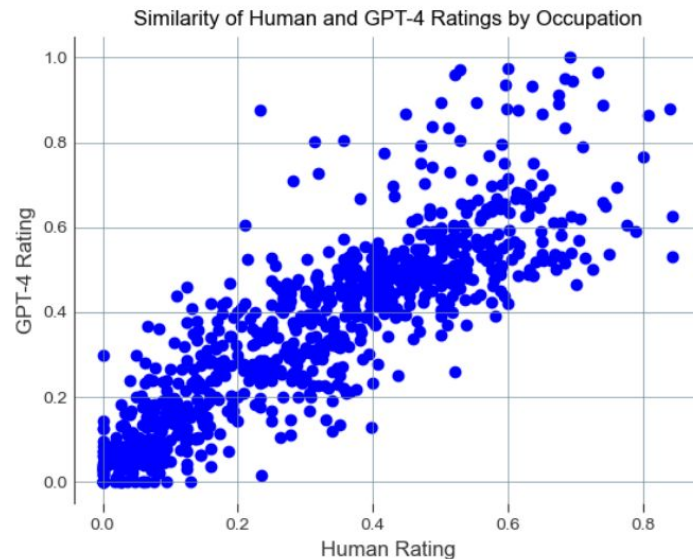
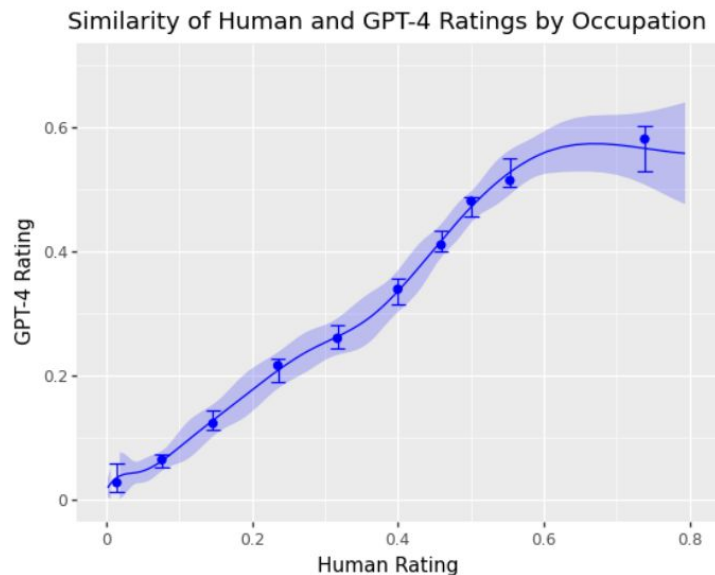


Figure 2: Human raters (x-axis) and GPT-4 ratings (y-axis) show a high degree of agreement about LLM exposure by occupation. We compute occupation-level exposure in these figures by averaging the task-level exposures under the β method. O*NET designates some tasks as "core" and others "supplemental". Core tasks are given twice the weight of supplemental tasks, and all weights sum to one. Near the highest levels of exposure following the β method of aggregating exposure scores to occupations, GPT-4 ratings tend to be lower than Human ratings. We present the raw scatter plot and the binscatter. Near the top end of exposure ratings, humans are on average more likely to rate an occupation as exposed.

Limitations of the Approach

Human labeling may be subjective

Task breakdowns may not capture all job complexity

Future AI capabilities uncertain

Results reflect potential, not guaranteed outcomes

Social and economic factors not fully included

Results - Overall Exposure Findings

~15% of tasks directly exposed to LLMs

30% exposed with supporting tools

50% exposure possible with advanced systems

80% of workers have some exposure

~19% of jobs have $\geq 50\%$ task exposure

Occupation Level Exposure				
	Human		GPT-4	
	mean	std	mean	std
α	0.14	0.14	0.14	0.16
β	0.30	0.21	0.34	0.22
ζ	0.46	0.30	0.55	0.34

Task Level Exposure				
	Human		GPT-4	
	mean	std	mean	std
α	0.15	0.36	0.14	0.35
β	0.31	0.37	0.35	0.35
ζ	0.47	0.50	0.56	0.50

Table 3: Summary statistics of our human and model exposure data. Tasks designated as core tasks for an occupation are given twice the weight as those indicated to be supplemental in the O*NET task file.

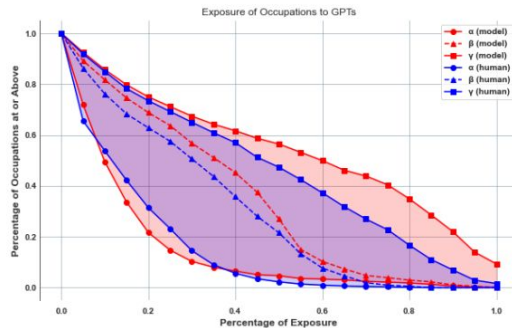


Figure 3: Exposure intensity across the economy, displayed in terms of percent of affected occupations. A given data point gives the percent of occupations with exposure below the given threshold. A previous version of this paper had two labels reversed in the chart, flipping human and model responses. In this figure, all tasks within an occupation are given equal weight.

Understanding Exposure (Important Clarification)

Exposure ≠ full automation

Means tasks can be done faster, not eliminated

Many tasks still require human oversight

Real impact depends on adoption and workflow changes

Results show **potential**, not outcomes

Variation Across Jobs and Skills

Higher-wage jobs tend to have higher exposure

Writing and programming tasks highly exposed

Science and critical thinking less exposed

Most jobs include both exposed and non-exposed tasks

Wide variation across occupations

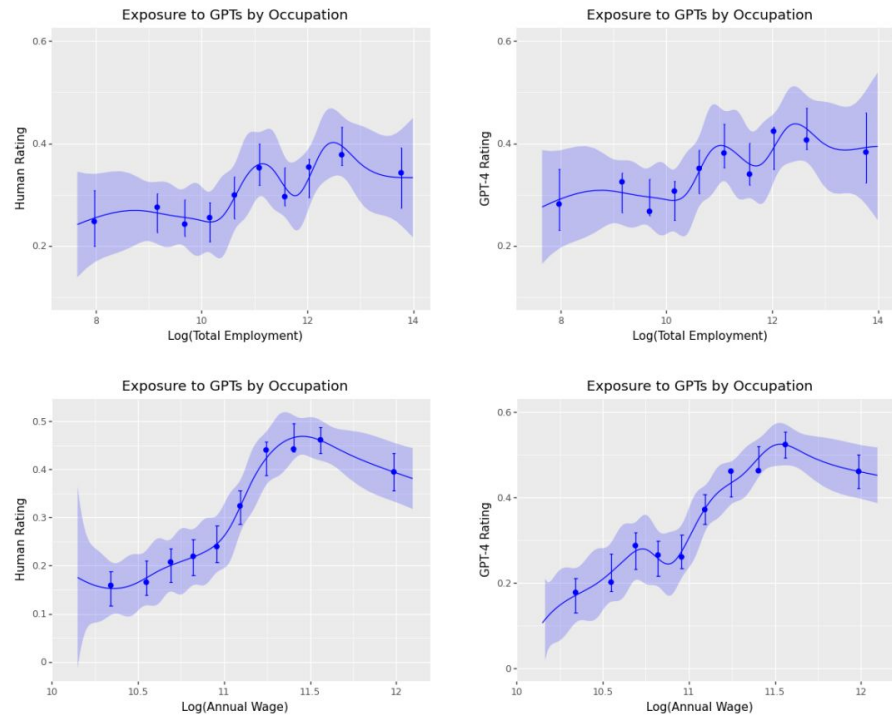


Figure 4: The binscatter plots depict the exposure to language models (LLMs) in various occupations, as assessed by both human evaluators and GPT-4. These plots compare the exposure to LLM and partial LLM-powered software (β) at the occupation level against the log of total employment within an occupation and log of the median annual wage for occupations. While some discrepancies exist, both human and GPT-4 assessments indicate that higher wage occupations tend to be more exposed to LLMs. Additionally, numerous lower wage occupations demonstrate high exposure based on our rubric. Core tasks receive twice the weight of supplemental tasks within occupations when calculating average exposure scores. Employment and wage data are sourced from the BLS-OES survey conducted in May 2021. In aggregating tasks to the occupation-level, we assign half the weight to O*NET supplemental tasks as we do for core tasks. All weights within an occupation sum to one.

Role of Education and Job Structure

Mid-to-high skill jobs show highest exposure

Degree-based jobs often more affected

Jobs requiring long training show lower exposure

Manual/physical jobs less affected

Industry differences reflect task types

Job Zone	Preparation Required	Education Required	Example Occupations	Median Income	Tot Emp (000s)	H α	M α	H β	M β	H ζ	M ζ
1	None or little (0-3 months)	High school diploma or GED (optional)	Food preparation workers, dishwashers, floor sanders	\$30,230	13,100	0.03	0.04	0.06	0.06	0.09	0.08
2	Some (3-12 months)	High school diploma	Orderlies, customer service representatives, tellers	\$38,215	73,962	0.07	0.12	0.16	0.20	0.24	0.27
3	Medium (1-2 years)	Vocational school, on-the-job training, or associate's degree	Electricians, barbers, medical assistants	\$54,815	37,881	0.11	0.14	0.26	0.32	0.41	0.51
4	Considerable (2-4 years)	Bachelor's degree	Database administrators, graphic designers, cost estimators	\$77,345	56,833	0.23	0.18	0.47	0.51	0.71	0.85
5	Extensive (4+ years)	Master's degree or higher	Pharmacists, lawyers, astronomers	\$81,980	21,221	0.23	0.13	0.43	0.45	0.63	0.76

Table 6: Mean exposure to GPTs by job zone. For each job zone, we also present the median of median annual income for each constituting occupation in USD, and the total number of workers in all occupations for that job zone, in the thousands. Task weights are equal for all tasks.

Group	Occupations with highest exposure	% Exposure
Human α	Interpreters and Translators	76.5
	Survey Researchers	75.0
	Poets, Lyricists and Creative Writers	68.8
	Animal Scientists	66.7
	Public Relations Specialists	66.7
Human β	Survey Researchers	84.4
	Writers and Authors	82.5
	Interpreters and Translators	82.4
	Public Relations Specialists	80.6
	Animal Scientists	77.8
Human ζ	Mathematicians	100.0
	Tax Preparers	100.0
	Financial Quantitative Analysts	100.0
	Writers and Authors	100.0
	Web and Digital Interface Designers	100.0
	<i>Humans labeled 15 occupations as "fully exposed."</i>	
Model α	Mathematicians	100.0
	Correspondence Clerks	95.2
	Blockchain Engineers	94.1
	Court Reporters and Simultaneous Captioners	92.9
	Proofreaders and Copy Markers	90.9
Model β	Mathematicians	100.0
	Blockchain Engineers	97.1
	Court Reporters and Simultaneous Captioners	96.4
	Proofreaders and Copy Markers	95.5
	Correspondence Clerks	95.2
Model ζ	Accountants and Auditors	100.0
	News Analysts, Reporters, and Journalists	100.0
	Legal Secretaries and Administrative Assistants	100.0
	Clinical Data Managers	100.0
	Climate Change Policy Analysts	100.0
	<i>The model labeled 86 occupations as "fully exposed."</i>	
Highest variance	Search Marketing Strategists	14.5
	Graphic Designers	13.4
	Investment Fund Managers	13.0
	Financial Managers	13.0
	Insurance Appraisers, Auto Damage	12.6

Table 4: Occupations with the highest exposure according to each measurement. The final row lists the occupations with the highest σ^2 value, indicating that they had the most variability in exposure scores. Exposure percentages indicate the share of an occupation’s task that are exposed to GPTs (α) or GPT-powered software (β and ζ), where exposure is defined as driving a reduction in time it takes to complete the task by at least 50% (see exposure rubric A.1). As such, occupations listed in this table are those where we estimate that GPTs and GPT-powered software are able to save workers a significant amount of time completing a large share of their tasks, but it does not necessarily suggest that their tasks can be fully automated by these technologies. All tasks are assigned equal weight within an occupation.

Role of Complementary Tools (Key Insight)

LLM-powered tools significantly increase exposure

Impact more than doubles when tools are included

Software systems amplify LLM capabilities

Real-world impact depends on integration

Suggests broader economic transformation

Occupation Level Exposure				
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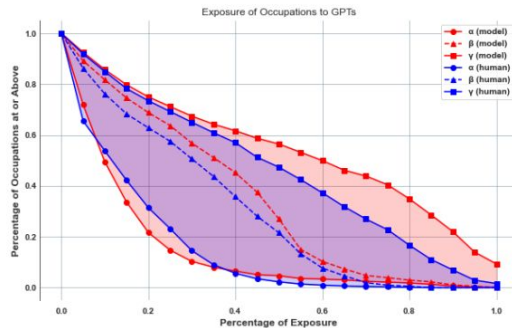


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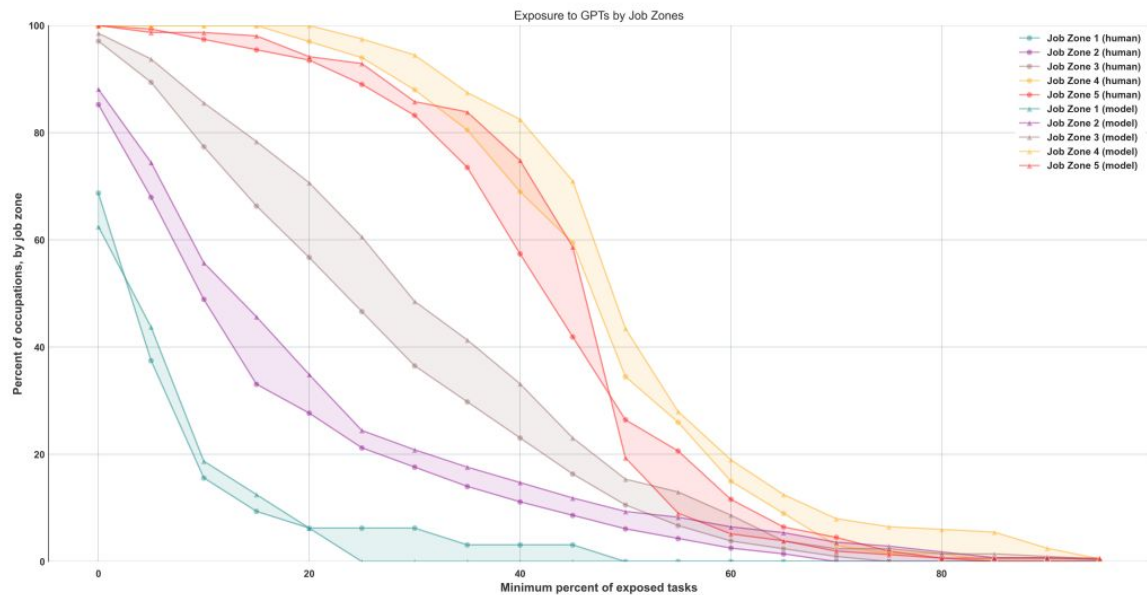


Figure 5: β exposure ratings of occupations in the five Job Zones, which are groups of similar occupations that are classified according to the level of education, experience, and on-the-job training needed to perform them. All tasks are weighted equally.

Key Takeaways from Results

LLMs have broad, economy-wide impact potential

Exposure depends on task type, not just job title

Higher-skill jobs are not protected

Complementary technologies are critical

Supports idea of LLMs as general-purpose technologies

Validation of Measures - Purpose of Validation

Compare results with prior AI/automation studies

Test consistency and reliability of new measures

Position LLM findings within existing research

Ensure credibility of approach

Validation Results

Strong positive correlation with prior AI exposure measures

Especially aligned with task-based methods (SML)

Confirms validity of the new rubric

Shows consistency with established research

	GPT-4 Exposure Rating 1		GPT-4 Exposure Rating 2		Human Exposure Rating	
	(1)	(2)	(3)	(4)	(5)	(6)
Software (Webb)	0.00113*** (0.00031)	0.00123*** (0.00031)	0.00111*** (0.00031)	0.00119*** (0.00031)	0.00096*** (0.00031)	0.00101*** (0.00031)
Robot (Webb)	-0.00378*** (0.00032)	-0.00405*** (0.00031)	-0.00377*** (0.00034)	-0.00399*** (0.00033)	-0.00371*** (0.00029)	-0.00383*** (0.00028)
AI (Webb)	0.00080*** (0.00030)	0.00090*** (0.00029)	0.00036 (0.00030)	0.00045 (0.00030)	0.00067** (0.00030)	0.00071** (0.00030)
Suitability for Machine Learning	0.29522*** (0.04503)	0.26888*** (0.04418)	0.28468*** (0.04404)	0.26245*** (0.04342)	0.19514*** (0.03990)	0.18373*** (0.03886)
Normalized Routine Cognitive	0.06601*** (0.00886)	0.06868*** (0.00894)	0.04743*** (0.00872)	0.05015*** (0.00879)	0.03568*** (0.00671)	0.03659*** (0.00669)
Normalized Routine Manual	-0.11147*** (0.00785)	-0.11371*** (0.00789)	-0.09390*** (0.00817)	-0.09561*** (0.00818)	-0.11045*** (0.00741)	-0.11152*** (0.00744)
AI Occupational Exposure Score	0.00993 (0.01107)	0.02465** (0.01059)	-0.01537 (0.01160)	-0.00265 (0.01114)	0.00630 (0.00918)	0.01252 (0.00845)
Frey & Osborne Automation	-0.03024* (0.01835)	-0.03950** (0.01841)	-0.00364 (0.02007)	-0.01217 (0.01972)	-0.03890** (0.01883)	-0.04253** (0.01858)
Log Avg. Salary	0.05804*** (0.01870)		0.04863*** (0.01860)		0.02531 (0.01727)	
Constant	-1.12937*** (0.26859)	-0.45743*** (0.15327)	-0.96117*** (0.26365)	-0.39935*** (0.15017)	-0.47078* (0.24684)	-0.17706 (0.13256)
N	680.00000	681.00000	680.00000	681.00000	680.00000	681.00000
R ²	0.68741	0.68212	0.60737	0.60198	0.71213	0.71126

Table 9: Regression of LLM-exposure scores on prior measures of occupational exposure to AI and automation. We also include annualized wages from the BLS-OES survey in May 2021. Each measure is kept in its original scale, with the exception of routine cognitive and routine manual scores from (Acemoglu and Autor, 2011a). Those two scores are standardized to mean zero and variance 1. Generally we find strong positive associations with previous efforts, though large residual variance to still be explained by our new measures. Columns 1 and 2 are based on our main β exposure measure from GPT-4 ratings. Columns 3 and 4 are based on a similar slightly different exposure rubric also rated by GPT-4 for robustness. Columns 5 and 6 reflect human ratings on the same rubric as columns 1 and 2. Occupation-level scores are built using the core/supplemental task weights, assigning supplemental tasks as having half the weight of core tasks.

Key Differences from Past Studies

Weak or negative correlation with robotics/manual measures

LLMs mainly affect cognitive and language tasks

Physical work remains less exposed

Highlights unique nature of LLM impact

New Insights from Validation

28 - 40% of variation not explained by prior models

Indicates LLMs introduce new patterns of impact

Earlier studies lacked insight into modern LLMs

Confirms need for updated measurement approaches

Discussion & Conclusion - LLMs as General Purpose Technology

Meet key criteria: improve, spread widely, enable innovation

Affect many industries and occupations

Impact grows through complementary tools

Similar to past transformative technologies

Adoption and Real-World Impact

Adoption depends on trust, cost, and usability

Simple interfaces can drive rapid adoption

Barriers: bias, errors, need for verification

Impact varies across industries and regulations

Path of Impact (Augmentation → Automation)

Short term: assist and enhance human work

Long term: some tasks may become automated

Jobs may change before they disappear

Transition may reshape job structures

Policy and Economic Implications

Risk of labor disruption and inequality

Need for education and worker retraining

Importance of social safety nets

Policymakers must prepare for change

Limitations and Future Research

Focus on U.S. limits global generalization

Does not fully capture future AI developments

Some capabilities (e.g., vision) not included

Future work: adoption patterns and real-world performance

Final Conclusions & Takeaways

LLMs have broad and growing impact on work

~19% of jobs may face high exposure

Higher-skill jobs are significantly affected

Real outcomes depend on adoption and policy

LLMs likely to shape the future of work